

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. Three conditions are separated: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared encoders of bounded sensitivity, and the bound grows as the square root of the population read: at the patch order of a simulated cortical sheet it constrains at most 10 sites, so on larger populations the link from coherence to content is an assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor. Lean 4 verifies these implications, and one model satisfies all connecting hypotheses together, which shows consistency, not cortical realization.

The conditions yield a disconfirmable comparison: at matched phase coherence and matched single-region decoding accuracy, decoders fitted independently on two regions should agree more, beyond their separate accuracies, on perceived than on unperceived stimuli classified by a no-report marker. The regions on which compatibility is assessed still need a selection principle, so the proposed correlate of unity awaits an operational definition. An exploratory EEG analysis is a measurement check only. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015) and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields; identifying them with perceptual content requires independent decoding evidence.

Binding-by-synchrony and communication-through-coherence accounts

17 treat phase relations as the medium of integration or of selective routing
 18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
 19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
 20 leave open what fact about synchronized activity makes it one content rather
 21 than coordinated activity carrying several. The contribution here is to sep-
 22 arate the three requirements and prove which follow from which. Phase
 23 order constrains shared content only through encoders related on overlaps,
 24 with a bound that grows as the square root of the population it is read
 25 on. Compatible local descriptions, not synchrony, determine a unique global
 26 state. A contracting self-map fixes one state but can reconstruct accurately
 27 only a family of bounded diameter. Each result names a measurement that
 28 phase data alone cannot supply.

29 The objection that synchrony does not fix content is older than this work.
 30 Critiques of the temporal binding hypothesis argued that synchrony, where
 31 it occurs, cannot say which features belong together or what they jointly
 32 represent: that grouping has to be computed before synchrony could sig-
 33 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
 34 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
 35 and a pigeonhole count. What formalization adds to the informal argument is
 36 quantitative and bookkeeping content. It gives the rate at which a coherence
 37 bound on content weakens with population size and the population at which
 38 it stops constraining anything, the diameter within which one contracting
 39 map can reconstruct, and the number of codes a separated family of states

40 requires. It also gives a complete inventory of the premises a synchrony-to-
41 content argument consumes, each of which names a measurement, together
42 with one model in which all of them hold at once.

43 The familiar noisy mean-field oscillator supplies a stationary coherence
44 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource
45 condition for a declared actuator, bounds that connect phase order to con-
46 tent only through specified encoders, and conditions under which compatible
47 descriptions glue and an accurate reconstruction fixes the glued state. A
48 Lean 4 development checks these implications and a joint witness satisfies
49 their eight connecting hypotheses in a specified model. It proves neither that
50 cortex satisfies those hypotheses nor that the glued state is experienced. The
51 latter identification motivates the framework and needs its own endpoint.

52 The empirical question is which measurements could support or discon-
53 firm the proposed connections. A phase resultant can miss a locally or-
54 dered winding field; stationary phase distributions can leave the coupling
55 mechanism unidentified; and coherent adaptation can lose alignment with
56 a designated content template. Independent decoders must identify shared
57 quantities before their agreement can be tested. We therefore distinguish sta-
58 tionary branches from finite observed onset, and compare prespecified field
59 and connectivity onset models on held-out emergence recordings. The com-
60 parison tests a specified spatial prediction, not the presence of experience.
61 Section 10.2 states the comparison that bears on conscious content directly.

62 Sections 2–5 develop the three requirements; Section 6 applies their mea-

63 surement obligations to a digital candidate. Section 7 states what the formal
64 composition establishes, and Sections 8–9 examine observables and the cor-
65 tical test. Table 1 lists the connecting hypotheses the article argues from;
66 Supplemental Material gives the full chain, derivations and controls. For a
67 theoretical article the Methods are the definitions, proofs and simulations.
68 Each result is stated in the section that uses it, Supplemental Material Ta-
69 ble S1 maps it to its Lean declaration, and every simulation’s model, param-
70 eters and controls are given in Supplemental Material. The Results are the
theorems of Sections 3–5 and the numerical controls of Section 8.

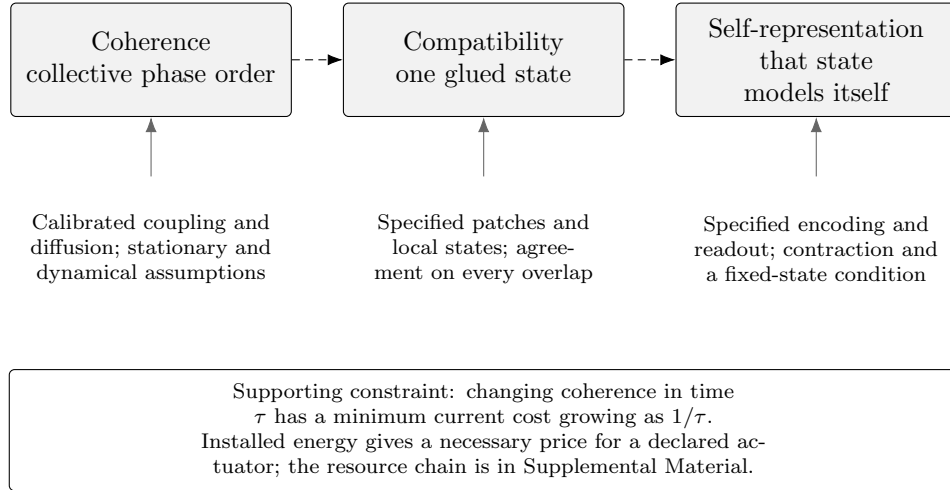


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the connecting hypotheses of the formal composition that the article argues from.

71

72 2 Limits of phase order

73 Several limits hold without the cortical hypothesis. Equal phases can coexist
 74 with incompatible local descriptions; phase order constrains shared content
 75 only through encoders related on overlaps. A winding field can have zero
 76 global resultant while remaining locally ordered. A constant readout can have
 77 a unique fixed point without reconstructing its input. A region encodes only
 78 the states it distinguishes, and by a declared deadline a site can report only
 79 changes inside its causal past. These limits reflect familiar phase oscillator
 80 and information constraints (Acebrón et al., 2005; Ott & Antonsen, 2008;
 81 Dörfler & Bullo, 2014; Breakspear, 2017). The sections below state each one
 82 beside the condition or measurement it motivates.

83 3 Coherence: physical coupling and its cost

84 3.1 Dynamics and the stationary threshold

85 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

86 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
 87 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
 88 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-

quency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

where I_n are modified Bessel functions and ψ is the collective phase. The formal derivation uses the absence of stationary current for a periodic gradient drift to obtain this density, then verifies normalization and self-consistency. At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz & Mirollo, 1991). Above threshold, exactly one positive solution coexists with zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero. These linear estimates concern the limiting equation. Relating them to finite

107 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 108 tion; exchange symmetry alone does not establish the needed independence
 109 (Sznitman, 1991).

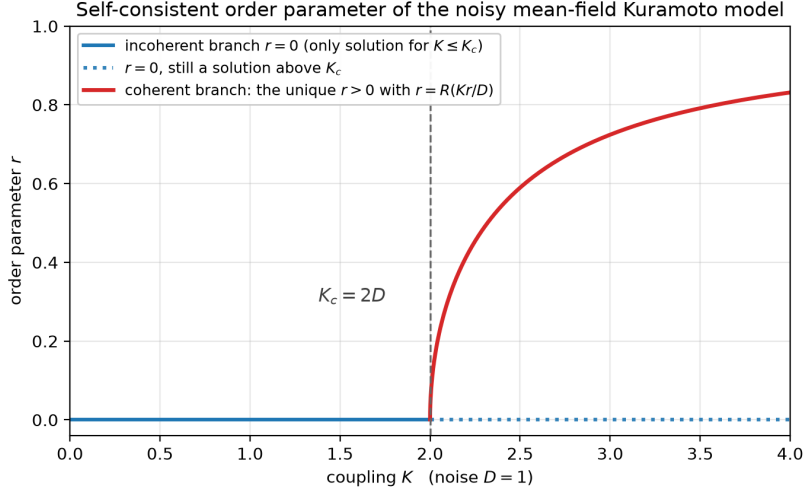


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

110 3.2 A speed limit on changes of coherence

111 One thermodynamic statement bears on coherence directly. For a smooth
 112 positive periodic phase density with probability current J and diffusion D ,
 113 the current functional $\sigma_J = \int J^2 / (D\rho) d\theta$ satisfies

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau} \quad (4)$$

114 while $0 < r < 1$. A change of coherence completed in time τ therefore has a
 115 minimum current cost that grows as $1/\tau$, whereas holding a stationary order
 116 has no positive floor. Read the other way, a current budget C sets a minimum
 117 duration $\tau \geq [\Delta \arcsin r]^2/(DC)$ for a given change. This bears on ignition-
 118 like transitions, in which large-scale coherence rises abruptly (Dehaene et al.,
 119 2014), and on recovery latency after anaesthesia: each has a floor set by the
 120 current available to drive it. The bound is a heat or power statement only
 121 when J is identified with a physical flux, a reservoir at a stated temperature
 122 is specified and local detailed balance holds for the steps concerned; only
 123 then does σ_J convert to entropy production (Landauer, 1961; Norton, 2005;
 124 Shenker, 2000). Installed energy supplies a separate, necessary price: for
 125 a declared actuator whose response modes carry installation prices, mean
 126 coupling is at most κU_{inst} for a mode-data factor κ , so a coherent stationary
 127 branch requires installed energy $U_{\text{inst}} > 2D/\kappa$, and no cortical value of κ
 128 is known. Supplemental Material gives both derivations, the register and
 129 feedback budgets, and the sensitivity of the metabolic comparison to its
 130 assumptions.

131 4 Compatibility: from coordinated activity 132 to a global description

133 4.1 Why common timing is insufficient

134 The formal setting is a sheaf: each patch of the substrate carries local de-
135 scriptions, and restriction maps cut a description on a patch down to any
136 smaller patch. A *section* is a description on one patch, and a *global section* is
137 a description on the whole substrate. For overlapping patches U and V , com-
138 patibility means that restricting either local description to $U \cap V$ gives the
139 same object. Descriptions can agree on shared quantities while contributing
140 different information elsewhere. Compatible sections of the specified sheaf
141 have a unique global extension. On a finite cover, local finite spatial mea-
142 sures agreeing as measures on every overlap also have a unique finite-measure
143 extension whose sheaf image is the glued section. No mass-one normalization
144 is imposed.

145 In the cup example, visual and reaching descriptions must express a
146 shared quantity, such as cup location, in common coordinates. Restriction
147 retains that quantity while dropping details particular to each description.
148 Agreement permits surface appearance and hand configuration to contribute
149 to one situation. The mathematical counterexample assigns the same phase
150 to every patch while two local mass profiles disagree on a shared site. Those
151 profiles have no common extension. Because they are chosen independently

152 of phase, synchrony places no constraint on their disagreement.

153 This distinction limits the biological interpretation. Receptive territories
154 and content domains must be related to spatial patches, and the descriptions
155 must support the required extension. Matching overlap marginals of de-
156 coded probability laws need not determine a unique global joint distribution
157 (Abramsky & Brandenburger, 2011).

158 The cover is itself a choice, and the construction constrains it in one way
159 only: synchronized patches agree where they overlap. Where they do not
160 overlap there is nothing to compare, so a cover by pairwise disjoint patches
161 satisfies that obligation for *every* assignment of local descriptions and phases
162 and the gluing theorem still returns a unique global section, which is the
163 tuple it was handed. Membership of the class then reports nothing about the
164 system beyond the patches covering it. What a gluing claim asks of a cover
165 is therefore a non-degeneracy the theorem does not assume: that distinct
166 patches meet, and that what they share is a quantity fixed independently of
167 the agreement it is being used to establish.

168 The cup example exposes a second gap, between two kinds of overlap.
169 The sheaf is built over a spatial substrate, so its patches overlap where they
170 share sites. Visual and motor territories that agree on cup location overlap
171 in what they describe, not in where they lie, and the construction supplies
172 no relation between the two. One route recasts the construction over content
173 variables: each region becomes a partial decoder whose domain is the set of
174 variables it estimates, and two regions overlap exactly where they estimate

175 a common variable. The gluing argument needs only a cover and restriction
 176 maps, so it applies formally to such a cover, but no result here is stated
 177 for one, and which variables and decoders form it is an empirical question.
 178 Until a principle selects the cover, the proposed correlate of unity has no
 179 operational definition: gluing on a declared cover is proved, and which cover
 180 cortex realizes is open.

181 4.2 When coherence does constrain content

182 A declared relation between phase and content gives a positive result. Let
 183 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 184 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 185 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance,
 186 meaning that an encoder's output moves by at most L times the chord
 187 between two input phases, and require $e_{i,x} = e_{j,x}$ wherever patches i and
 188 j overlap. Encoders may vary across sites; their equality on overlaps is a
 189 supplied relation between descriptions of the shared quantity. Write $z_k = e^{i\theta_k}$
 190 and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The phasors' squared deviations from their
 191 mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$, and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$
 192 gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair. Consequently every overlap
 193 discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1 - r^2)}. \quad (5)$$

194 Here discrepancy is the norm of the difference, and r and N are read on
 195 whichever population the bound is applied to. That choice is what decides
 196 whether the estimate says anything. A patch carries an order parameter
 197 of its own, and reading the bound there carries that patch's resultant and
 198 that patch's site count, so a large population observed through small patches
 199 is constrained by the small count. Read instead on the whole array, the
 200 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
 201 from an average over N sites. The order cannot be improved in general:
 202 one oscillator in antiphase among N leaves $N(1 - r^2)$ near 4. At $r = 1$ the
 203 observations agree exactly at either scale. A shared scalar encoder is a special
 204 case. Dropping the overlap relation permits disagreement even at perfect
 205 locking. A common nonexpansive linear projection preserves agreement, but
 206 agreement of a projection does not establish agreement of the full vector.

207 What Eq. (5) bounds is a chord, which never exceeds 2, so the estimate
 208 constrains anything only where the right-hand side falls below that. At the
 209 sheet's own patch order, the bound is informative for at most 10 sites; beyond
 210 that, the coherence-to-content link requires the bridge assumption E78 of
 211 the cortical hypothesis (Table 1 gives the full composition and its edges).
 212 Supplemental Material provides the detailed derivation of this saturation,
 213 and states three conditional derivations that replace the uniform factor $\sqrt{N}$
 214 by a count of disagreeing pairs, a chain across a cover's shared sites, or a
 215 spectral gap at a frequency-locked configuration.

216 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$

217 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 218 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 219 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (6)$$

220 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 221 duce an initial mismatch. Approximate observations leave a nondecaying
 222 term in the error bound. Taking a uniform bound on Eq. (5) connects this
 223 update to the phase process.

224 **4.3 What approximate agreement determines**

225 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 226 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 227 Weights that vanish outside their patches and sum to one at each site select
 228 a global profile by weighted averaging. The selection lies within ε of each
 229 patch there, and any two such selections differ by at most ε , independently
 230 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 231 $L\varepsilon$ across selections.

232 At zero tolerance, the selection is the unique exact extension. At every
 233 positive tolerance there are families with no exact extension; selection then
 234 gives an approximation whose dependence on weights is bounded. Supple-
 235 mental Material gives a three-site witness attaining the bound. The mass-
 236 profile construction uses the scalar case of the content dynamics and addi-

tionally requires nonnegativity of the initial profiles and observations. The general vector result supplies no identification with the spatial-measure sheaf. These constructions concern declared objects: a specified cover, contents valued in a normed space, and encoders related on overlaps. They select neither the neural cover nor the decoder, and identifying their contents with perceptual content, or establishing that the identification bears on experience, is a further modelling task; Section 8 sets out what a measurement of it would have to establish.

5 Self-representation: a state that models itself

5.1 The reflexive construction

Compatibility gives a global state. Reflexivity relates it to an encoding within that state. Equip global sections with a complete metric. *Restriction resonance* is the additional condition that the encoding is the state's own restriction to a region $\mathcal{A}$, so that the region encodes the state by being part of it. Under that condition the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \tag{7}$$

253 The region and readout are fixed across the state space. If F contracts and
 254 fixes the section s_* formed from the local descriptions, Banach's theorem
 255 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 256 (Table 1 gives the full composition and its edges) supplies the contraction
 257 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
 258 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for
 259 a relaxation time $\tau > 0$. The factor is stipulated, not derived. It borrows
 260 the rate $(K - 2D)/2$ at which the first harmonic of the uniform density
 261 grows (Section 3), a rate about a different state from the one F acts on,
 262 and no result derives a Lipschitz law for an encoding or readout from the
 263 phase dynamics. Constructing a map with that factor remains a modelling
 264 obligation. A fixed contraction can reconstruct accurately only a restricted
 265 family of states: if its error is at most ε throughout that family, any two
 266 members are at most $2\varepsilon/(1 - \Lambda)$ apart.

267 In the cup example, the global state includes the represented scene and the
 268 perceiver's relation to it. A local encoding must retain enough information
 269 for the readout to recover that modelled state; cup position alone would be
 270 insufficient if hand configuration and viewpoint could not be reconstructed.

271 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 272 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (8)$$

273 At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for

fixed positive ε and τ . Exact uniqueness survives while the error guarantee weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (9)$$

Thus M relevant states pairwise more than 2ε apart require at least M distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$. Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the existing self-prediction map.

This is an additional accuracy criterion on a family, with no requirement for multiple exact fixed points. For the same fixed contracting map used by E89, however, the diameter bound above is an additional restriction: at zero error the family contains at most one metric-distinct state. A broad family therefore requires a different, input-conditioned reconstruction map, one map F_u with its own fixed point for each input u , with its own accuracy and stability assumptions. No such map is formalized or constructed here. Under E89 the proposed minimal-self correlate is therefore a condition on one self-state, not a model of a self that follows a changing scene. A blind

293 encoder or constant readout fails the stated criterion for separated states
 294 despite having a unique fixed point.

295 Where the encoding is read as the state’s own restriction to a region, the
 296 criterion becomes a condition on the region. Restriction resonance requires
 297 the encoder and that restriction to identify the same pairs of states, so a
 298 boundary reconstructing two relevant states more than 2ε apart while its
 299 region reads them identically is not resonant. Read forwards, a resonant
 300 boundary that reconstructs a pairwise separated family restricts injectively
 301 on it. Which states a region reads identically is fixed by where the region
 302 sits, before an encoding is chosen, so the resolution the family demands is
 303 the region’s and no readout supplies it. The implication runs one way: a
 304 region that does resolve the family is not thereby shown to carry a resonant
 305 encoding, and E89 supplies no such resolution.

306 Resonance and resolution are jointly satisfiable without a local fold. Take
 307 the region to be the whole substrate and use the identity readout: restric-
 308 tion is the identity, so this boundary is resonant and reconstructs every de-
 309 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 310 E89’s strict contraction premise on a nontrivial metric space. A properness
 311 condition on the region is therefore a separate requirement, and the fixed-
 312 contraction diameter bound remains in force when E89 is also required. With
 313 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 314 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 315 so the family is no larger than the δ -packing number of what the region car-

316 ries. That is the shape a continuous family of states needs, and at the whole
 317 substrate it says nothing, because the region's range is everything. The gain
 318 L is a declared input, derived here for no physical readout, and the bound
 319 weakens without limit as the gain grows.

320 **5.3 Agreement after a change takes communication**

321 Snapshot compatibility does not specify how quickly descriptions respond
 322 when a shared quantity changes. In a synchronous network whose updates
 323 read only the site's own state and declared neighbours' messages, two initial
 324 configurations agreeing on a site's causal past through T rounds produce the
 325 same state there at round T . The deadline consequence is the one Section 2
 326 states, and it follows from the reconstruction bound in Section 5.2: a report
 327 from that site is a readout of a code that the two interventions leave identical.

328 A three-site delay line witnesses failure after one round and exact report-
 329 ing after two; cutting the path prevents delivery at every round. A report
 330 may still happen to be correct about one fixed world. The bound concerns
 331 responsiveness across interventions.

332 Throughout this section *prediction* names a reconstruction map, and the
 333 relevant family of states, the metric, the tolerance, the communication graph
 334 and the deadline are all declared inputs, as are the represented variables and
 335 the fidelity counted as accurate. Reading the map as temporal neural pre-
 336 diction would require dynamics; reading the resulting code count as physical
 337 memory capacity would require a hardware model, which a state-dependent

338 readout would have to extend.

339 6 What the conditions ask of a digital candi- 340 date

341 A claim that a digital system, such as a large language model (LLM) run-
342 ning on graphics processing units (GPUs), has unified content or represents
343 itself commits its maker to the measurements this account asks of cortex.
344 The maker must identify the shared quantities the system’s parts describe,
345 show that decoders fitted independently on those parts agree about them,
346 and show that a declared local code separates the relevant family of states
347 and survives intervention. The indicator-properties approach assesses arti-
348 ficial systems against computational properties drawn from theories of con-
349 sciousness (Butlin et al., 2023); the conditions here are fewer and each is
350 a measurement rather than an architectural property. A digital system is
351 also where they can be measured today, with full access to its state and
352 exact interventions on it: fitting decoders for one shared quantity on two
353 separate layers, heads or token positions, and testing their agreement and
354 whether reconstruction survives intervention, would carry out the proposed
355 measurement directly.

356 One constraint follows from the architecture alone. Within one forward
357 pass of a causally masked model, position t reads only its causal past, so an
358 intervention outside that past cannot be reconstructed at t by that step. This

359 is the deadline result of Section 5 applied to the mask, and it bounds which
 360 variables a position can encode about the rest of the sequence within one
 361 step. A second constraint needs a premise the architecture does not supply:
 362 a linear encoder and readout factoring through d coordinates pay reconstruc-
 363 tion error at least the discarded spectral tail of the coupling operator, but
 364 query/key dimension does not supply such a bottleneck for softmax atten-
 365 tion. These results rank no architecture family, imply no general inferiority of
 366 GPU hardware and settle no question of machine experience. Supplemental
 367 Material gives the bottleneck derivation, a synthetic two-view task with its
 368 controls, the information ceiling of token outputs and the hardware scope.

369 **7 The formal composition**

370 The Lean development composes the three requirements into one conditional
 371 theorem. Its passive and feedback branches each accept eight connecting hy-
 372 potheses, running from finite capacity through a register, dissipation and a
 373 continuum coupling to the field, coherence, the cover and the self-prediction
 374 fixed point. Each edge is supplied rather than derived from the one above it,
 375 and the inferential content of the composition is its last step, where contrac-
 376 tion and a fixed glued section give uniqueness. Table 1 lists the four edges the
 377 article argues from; Supplemental Material gives the full chain, its thermo-
 378 dynamic edges and the register ledger of the feedback branch. Both branches
 379 also take a cover carrying exact overlap compatibility and a reflexive bound-

380 ary specifying encoding and readout, and conclude that the glued section is
 381 the unique fixed point of self-prediction. Composing the edges establishes
 382 that the premises are consistent, named in one place and machine-checked
 383 together; it does not establish that self-representation follows from coher-
 384 ence. Joint witnesses satisfy every edge of each branch in a specified model,
 385 and establish no common cortical mechanism. The development declares no
 386 additional axioms beyond its libraries’ foundational axioms.

Table 1: The four connecting hypotheses the article argues from, shared by both branches of the formal composition. Supplemental Material gives the full eight-edge chain and, in Table S1, maps the node-level results.

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

387 8 What the observations establish

388 The composition holds under its premises. Whether observation can estab-
389 lish those premises is a separate question, and the three requirements it
390 connects concern different objects: phase dynamics, local descriptions and
391 internal encodings. Their observables must retain these distinctions. The

392 following controls show which mechanisms remain indistinguishable, which
 393 temporal approximations fail, which states one observable cannot separate,
 394 and what independent content measurements must establish. Each is a finite
 395 numerical regime on a declared model, with full parameters and controls in
 396 Supplemental Material, and none of them calibrates a cortical quantity.

397 **8.1 Stationary distributions leave the mechanism un-** 398 **determined**

399 Equation (2) gives a concentration–coherence relation with no fitted shape
 400 parameter. Estimate r from circular resultant length and a from density
 401 shape: inverting R from the measured r would make agreement automatic.
 402 Even independently estimated agreement identifies a statistical family shared
 403 by different mechanisms.

404 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
 405 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
 406 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 407 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
 408 and diffusion D_0 have that same density. Their stationary distributions co-
 409 incide window by window despite different causes.

410 Resolved dynamics supply additional information. In the ideal model
 411 with continuous latent paths, quadratic variation accumulates at rate $2D$
 412 and conditional increment means constrain drift; a filtered observation of

413 the phase needs its own observation model before either is read off. Separat-
 414 ing endogenous coupling from a matched common drive additionally requires
 415 a measured drive or controlled perturbation; identifying electromagnetic cou-
 416 pling requires evidence specific to that interaction. Supplemental Material
 417 gives the equivalence construction and its numerical check. Phase statistics
 418 pooled over time and over bipolar pairs cannot supply these estimates: on
 419 public sedation EEG (Bajwa et al., 2025), the pooled pipeline places a per-
 420 fectly phase-locked signal at $r = 0.0001$, so a spatial test needs an estimand
 421 that removes the common rotation and a montage whose effect on spatial
 422 phase is modelled; Supplemental Material reports this estimator exercise and
 423 the dependence-aware design.

424 **8.2 Finite ramps change the observable onset**

425 The square-root law in Eq. (3) describes stationary solutions, and a chang-
 426 ing coupling needs time for the phase distribution to approach them. In a
 427 deterministic mean-field control crossing the threshold, none of the scanned
 428 rates meets the declared tracking tolerance at onset, so a slow control vari-
 429 able supplies no general guarantee of branch tracking. In a finite-population
 430 ramp study ($N = 2000$, $D = 1$, 32 replicas, 8 speeds), faster ramps de-
 431 lay escape. The fitted delay exponent 0.444 has a replica-bootstrap interval
 432 $[0.398, 0.496]$ that excludes the ideal $1/2$, a deterministic reference shows that
 433 the escape criterion changes the fitted exponent, and no common asymp-
 434 totic exponent is established. Delayed passage through a noisy bifurcation

(Berglund & Gentz, 2002) makes the apparent exponent depend on ramp speed and fitting window. Supplemental Material gives the sweep and its figure, the fluctuating-coupling and population-size controls, and the signed-network crossing bracket.

The ideal delayed-onset calculation motivates a square-root scaling hypothesis; its exponent is generic to suitable bifurcations and does not identify the coupling medium. A within-subject emergence design can vary the propofol taper and test how an operational onset changes with that manipulation, but interpreting the result as a coupling-rate law requires a measured or calibrated relation between taper and $\dot{K}$. The escape criterion must be set relative to each recording's pre-threshold fluctuation scale. Section 8.1 describes how D could be estimated from resolved phase traces.

8.3 A winding state leaves the order parameter ambiguous

The global order parameter r discards spatial arrangement. On a periodic sheet, a phase field can wind once around the sheet and have $r = 0$ while neighbouring sites remain locked. At identical frequencies, every such winding is stationary for an isotropic symmetric kernel: contributions at opposite separations cancel. It violates the quarter-turn initial-spread premise of the convergence result, which is why that result can still select the uniform state.

In the spatial control, the winding persists below the interpolated de-

456 cay length 0.1871 mm and unwinds above it (Fig. 3). At 0.1 mm, uniform
 457 and winding starts under the same coupling, diffusion and kernel end at
 458 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
 459 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 460 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 461 realizations. These are finite controls, not a cortical wave identification or a
 462 derived transition.

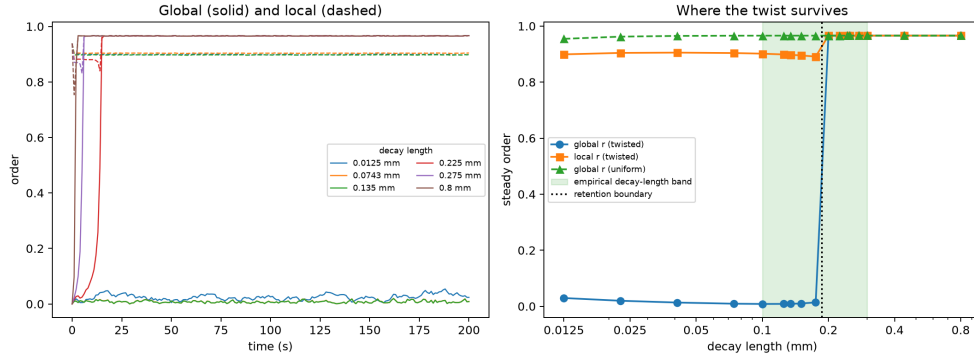


Figure 3: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

463 Linearization about a winding reweights each separation's coupling by
 464 the cosine of its phase advance. The resulting symmetric kernel has explicit
 465 mode rates: nonnegative reweighted coupling gives no growing mode, while

466 a positive rate destabilizes that winding. The criterion concerns linear rates,
 467 not asymptotic stability, and the measured retention boundary need not
 468 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$
 469 0 , where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 470 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 471 winding from that model even though the phase-only equations retain it as
 472 a stationary state.

473 Neither model turns these states into measured cortical spirals. The pe-
 474 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 475 no propagation speed, and the amplitude variable has no calibrated relation
 476 to a recorded envelope. Supplemental Material gives the spectral, amplitude
 477 and defect results. Locally, Eq. (5) can still bound content disagreement on a
 478 winding patch when shared encoders are supplied, although its global $r = 0$
 479 form is uninformative. A nonzero ring winding number q also obstructs a
 480 global real-valued phase lift; it has no identified content meaning here. Thus
 481 an onset criterion based only on r can register a change of basin as well
 482 as a change of coupling. The protocol of Section 9.3 needs a spatial phase
 483 observable.

484 8.4 Coherent adaptation can lose environmental align- 485 ment

486 An adaptive coupling rule might supply representational structure, but its
487 objective must reward the relevant agreement. For symmetric coupling, pair-
488 wise sine terms cancel in the population mean drift. Minimizing squared drift
489 at that fixed mean therefore rewards equalizing drift across nodes. In the
490 frequency-cluster control, this favors coupling between unlike-frequency clus-
491 ters, whereas the diagnostic environmental template rewards coupling within
492 clusters.

493 The formalization proves descent of the squared-drift objective for a spec-
494 ified gradient flow at fixed phases. Its functional differs from thermodynamic
495 entropy production: at a coherent stationary density the probability current
496 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
497 rial gives the operator, its gradient and the conditions of the descent result.

498 The finite control co-evolves noisy phases and symmetric, nonnegative
499 couplings at fixed total resource. Its template is a diagnostic covariance
500 of latent signals, never a training input. Across three illustrative seeds,
501 second-half order remains between 0.9167 and 0.9222, while the within-
502 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
503 0.6135 (Fig. 4). Random-update, permuted-gradient and frozen-kernel con-
504 trols are reported in Supplemental Material.

505 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on synthetic data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed con-

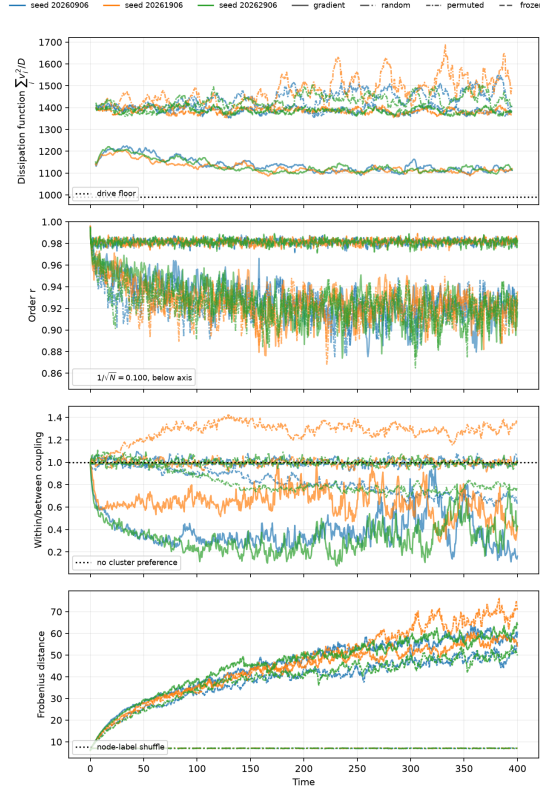


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

528 tent, phase carries a share 1.000 of the content entropy when content is built
529 from it and 0.000 when it is not. A phase-derived statistic reports com-
530 patibility on phase-locked patches whose local descriptions disagree. The
531 implemented content statistic detects their incompatibility. These controls
532 validate a measurement on known constructions; a neural content model and
533 decoder remain to be established.

534 Choosing the cover also chooses a candidate subject: trivial descriptions
535 can agree while carrying little content, and a partition can conceal disagree-
536 ment. The useful choice must therefore be constrained by what neural ac-
537 tivity encodes. The exact composition assumes compatibility of its supplied
538 cover. The content mechanisms above identify additional hypotheses under
539 which agreement can arise; they leave the empirical choice of content and
540 the connection to that cover explicit.

541 **9 The cortical hypothesis and an awakening**

542 **test**

543 The cortical proposal identifies a physical source of coupling and asks whether
544 its changes help explain recovery. The tests in Section 8 determine what
545 observations must distinguish before a phase trace can support this identifi-
546 cation.

547 9.1 Physical identification and calibration

548 The cortical hypothesis identifies part of effective coupling with endogenous
549 extracellular electromagnetic interactions. Fields can modulate spike timing
550 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
551 propose coordination of distributed neural information (Pinotsis & Miller,
552 2022; McFadden, 2020). Phase organization under propofol motivates study-
553 ing this regime (Bastos et al., 2021), but does not measure coupling.

554 The formal field predicate requires a jointly continuous kernel, a normal-
555 ized substrate, coupling equal to the kernel average, and no single site pro-
556 viding more than half that strength. Toy constructions satisfy it. A cortical
557 realization must specify the scale at which the kernel and its reduction ap-
558 ply: calcium imaging reports different state-dependent modularity patterns
559 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

560 Cortical columns are the obvious candidate unit, and they answer one
561 of the two demands the formalism makes. As the oscillator index they are
562 a serious proposal: they are delineated by response properties and anatomy
563 rather than by the agreement they would be used to establish (Mountcastle,
564 1997), and a phase can be estimated from a column-sized territory at the
565 electrode spacings used for cortical wave measurements (Townsend et al.,
566 2015). As the content cover they are a poor one, for three separate reasons.
567 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
568 the compatibility obligation without constraining anything (Section 4). The
569 cover condition additionally requires the patches to exhaust the substrate,

570 while much of association cortex has no accepted columnar parcellation. And
571 the column's standing as a canonical cortical unit is contested (Horton &
572 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
573 supplies the oscillator index; the content cover remains an open empirical
574 question, which the bridge assumption E78 records.

575 The spatial reduction is also testable. A row-sum aggregation at zero
576 diagonal tracks the uniform-coupling control over the tested spatial decay
577 range, while frequency heterogeneity displaces the operational threshold. The
578 fit window biases both estimates relative to $2D$; agreement with the control
579 is therefore the supported conclusion. Supplemental Material reports the
580 reduction, its numerical error and the distinct assumptions needed to identify
581 the kernel average with the scalar coupling of the installed-energy condition.

582 The field commitment requires physical evidence beyond a choice of math-
583 ematical representation, and the available support comparisons do not supply
584 it. A finite matrix spread over positive-mass cells preserves its total cou-
585 pling and phase correlation exactly, while point-supported vanishing under
586 an atomless measure and a disadvantage from excluding a better-correlated
587 pair concern specified support constraints; Supplemental Material reads the
588 same comparisons against a digital candidate. Neither supplies energy prices.

589 A spike-time shift alone supplies no coupling rate. The dimensionless
590 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and

591 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (10)$$

592 This rate describes how induced timing shifts accumulate into drift. Nei-
593 ther it nor the installation factor κ follows from a heat bound. Testing the
594 physical threshold requires independent estimates of the interaction kernel,
595 its aggregation and diffusion in the same cortical regime.

596 **9.2 Extracellular geometry and recovery**

597 The proposed control variable is extracellular geometry. Extracellular vol-
598 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
599 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
600 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
601 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
602 an effective-medium description predicts that a larger extracellular fraction
603 raises conductivity and reduces field amplitude.

604 The coupling-gated hypothesis consequently predicts that extracellular
605 contraction on awakening raises E and, holding the other factors in Eq. (10)
606 fixed, raises effective coupling. This argument supplies a direction; the
607 geometry–coupling relation, its magnitude and its role in human recovery
608 require measurement.

609 For a smooth crossing at nonzero speed, evaluating the stationary branch

610 along the changing control gives a deterministic tracking approximation that
 611 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (11)$$

612 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
 613 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
 614 proximation under the tested initialization. A separation of geometry and
 615 phase time scales can support tracking away from threshold. The awakening
 616 hypothesis needs a calibrated dynamical prediction for an operational onset
 617 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
 618 recovery latency.

619 **9.3 A protocol for the awakening experiment**

620 Neural inertia describes a behavioural barrier and hysteresis between anaes-
 621 thetic induction and emergence (Friedman et al., 2010); recovery can pass
 622 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
 623 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
 624 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
 625 mechanism from a synaptic one. A possible discriminator is spatial onset
 626 relative to independently estimated local field coupling and network connec-
 627 tivity. The present mean-field model has no spatially varying κ and derives
 628 no nucleation order, so a field-based onset map needs a separate model spec-

629 ified before the observations are inspected.

630 For that comparison, partition the recording into prespecified regions i
631 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
632 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
633 and a connectivity score c_i from a separately estimated structural network.
634 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
635 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic
636 approximation predicts

$$\hat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (12)$$

637 Predicted onset is the first crossing of a region-specific threshold fixed from
638 pre-onset fluctuations. Fit the two scales on training sessions, then com-
639 pare held-out onset times and site order under the same observation model.
640 Larger field scores predict earlier onset only when baseline order and diffu-
641 sion do not reverse the ranking. This declared approximation omits interre-
642 gional coupling and saturation; a successful ranking would support its field
643 covariate, not prove ephaptic causation. Supplemental Material specifies the
644 controls and failure conditions.

645 A within-subject protocol can manipulate emergence rate, with spatially
646 resolved recordings to compare the proposed field and connectivity models:

647 **1. Manipulate emergence rate.** Vary the propofol infusion taper
648 across sessions within the same subject, recording drug concentrations

649 or a validated pharmacokinetic proxy to quantify the manipulation.

650 **2. Measure behavioural recovery.** Record time-to-response to a stan-
651 dard stimulus as the independent behavioural endpoint, rather than
652 estimating ΔK directly.

653 **3. Resolve spatial onset.** Identify anatomical locations where coher-
654 ence first crosses the baseline-scaled threshold and score the two maps
655 of Eq. (12) on held-out sessions. The two covariate maps are fixed
656 before the outcome sessions are inspected.

657 **4. Match physical controls.** Confirm that the observed coherence-to-
658 amplitude relation holds under the specific local diffusion D and ob-
659 servation noise. The model’s quadratic-variation rate $2D$ applies to
660 the latent phase; a band-limited analytic phase is smooth below its
661 correlation time, and on a constructed field its squared-increment rate
662 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
663 fore requires an explicit observation model for filtering and sampling.

664 Predeclare the recording window, regional fluctuation thresholds, and a
665 shared timing-error model. Fit the two scales and that observation model
666 on training sessions only; on held-out sessions, score the observed site-onset
667 times and sites that never cross by the same regional predictive rule. Report
668 site order as a secondary outcome and behavioural time-to-response sepa-
669 rately. Sessions without a crossing are censored observations, not zero delays
670 or exclusions. Compare the two scores only where independently estimated

671 field and connectivity maps predict distinguishable onset patterns at the
672 recording resolution. A held-out field-model advantage would support that
673 specified spatial predictor; it would not identify an ephaptic cause, shared
674 content or experience. Supplemental Material gives the controls and pilot
675 quantities needed to judge whether the comparison is informative.

676 **10 Discussion**

677 **10.1 Unity and a minimal self as interpretive motiva-** 678 **tion**

679 We propose the state glued from compatible local descriptions as a formal
680 correlate of conscious unity, and its accurate internal reconstruction as a cor-
681 relate of a minimal self. In the cup example, this would mean one description
682 that preserves the cup's relation to the perceiver as well as the scene. Com-
683 patibility permits distinct contents and uncertainty; the shared quantities
684 must be chosen from independent evidence about what each region encodes.
685 The reconstruction need not be a verbal or autobiographical account.

686 The minimal self, in Gallagher's sense, is the pre-reflective sense of be-
687 ing the subject of an experience, present before any narrative or reflective
688 self-description (Gallagher, 2000). Two of its features motivate the proposed
689 correlate. It is not a separate object of experience but a perspective built
690 into the experienced scene; correspondingly, the reconstruction is not a sec-

691 ond representation beside the glued state but a map whose fixed point is that
 692 state, read through a part of it. It is immediate rather than inferred from
 693 a higher-order state; correspondingly, the fixed point needs no further state
 694 that represents the first. The correspondence is structural and does not show
 695 that a fixed point is experienced as a perspective. Self-model theory locates
 696 the phenomenal self in a transparent model a system has of itself (Met-
 697 zinger, 2003), and interoceptive-inference accounts ground it in prediction of
 698 the body’s own states (Seth, 2013). The present condition is narrower than
 699 either: it asks only that a declared local encoding recover the global state,
 700 including perceiver-relative variables, at a declared fidelity, and it says noth-
 701 ing about transparency or about which bodily variables must be modelled.
 702 Either account could supply the family of states and the fidelity that the con-
 703 dition leaves as inputs. Under E89 that family is one state at zero error and
 704 a family of diameter at most $2\varepsilon/(1 - \Lambda)$ within error ε (Section 5.2), so the
 705 condition as proved constrains a single self-state; a self that persists through
 706 change needs the input-conditioned map that is not constructed here. Its
 707 proposed relation to experienced perspective remains an interpretation. Nei-
 708 ther identification is among the connecting hypotheses of the composition
 709 (Table 1), and satisfying them would not measure experience. The unity
 710 proposal also lacks an operational definition. The gluing theorem holds for
 711 whatever cover it is given, a disjoint cover satisfies it trivially, and the cover of
 712 interest overlaps in content rather than in space (Section 4); until a principle
 713 fixes that cover, the correlate is a schema awaiting one.

714 10.2 A comparison that could disconfirm the link to 715 content

716 The framework predicts that conscious content tracks compatibility, not co-
717 herence. The comparison must also separate compatibility from how well
718 each region encodes the stimulus, because two accurate decoders agree auto-
719 matically. If decoders d_A and d_B estimate a shared stimulus quantity x with
720 errors $e_A = d_A - x$ and $e_B = d_B - x$, then

$$|d_A - d_B| \leq |e_A| + |e_B|, \quad |d_A - d_B|^2 = |e_A|^2 + |e_B|^2 - 2\langle e_A, e_B \rangle. \quad (13)$$

721 Consciously perceived stimuli are usually decoded better, so higher agreement
722 on those trials would follow from single-region fidelity alone. At matched
723 single-region error, agreement differs only through the covariance of the two
724 errors, which is the term compatibility bears on: two regions describing one
725 content should err together.

726 Present matched stimuli that are perceived on some trials and not on
727 others, for example at masking or rivalry threshold, and fit decoders for
728 one shared stimulus quantity independently on two regions that both encode
729 it. Classify trials by perceptual state with a no-report measure, such as
730 reflexive eye movements or pupil responses that track the dominant percept
731 in rivalry (Tsuchiya et al., 2015), so that the classification does not also
732 select for the neural correlates of reporting; report-based classification is a
733 secondary analysis. Match the two classes on phase coherence between the

regions and on each region’s own decoding error (or, where the stimulus value is uncertain, on each region’s signal strength). The statistic is the excess agreement over a fidelity-matched null: pair region *A*’s decode on one trial with region *B*’s decode on another trial of the same stimulus and error stratum, which preserves both marginal accuracies and removes any coupling of the errors.

The proposed link predicts a larger excess on perceived trials at matched coherence and matched single-region accuracy (Fig. 5). Equal excess across classes, or excess that follows coherence and not perceptual state, counts against it. A positive result is not yet specific: noise shared upstream of both regions also correlates their errors, so the excess should be compared against an upstream region’s decoding error, matched in the same way. A self-model comparison has the same form: at matched accuracy for the scene variable, the reconstruction of perceiver-relative variables, such as hand position relative to the target, from a declared local encoding should be more accurate on perceived trials. These are correlational tests of proposed correlates, and the no-report classification is itself an indirect marker of experience.

10.3 What a test of the full proposal must establish

A test needs separate estimates of effective coupling and diffusion, shared content on overlapping regions, and reconstruction from a specified encoding at a declared fidelity. The cover, relevant state family and decoder must be fixed without selecting them for agreement. An independent experiential

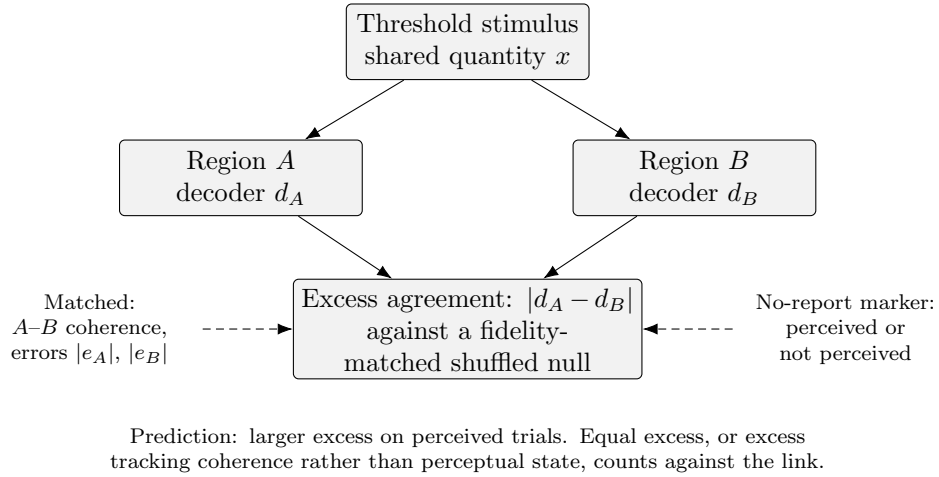


Figure 5: Design of the disconfirming comparison. Decoders for one shared stimulus quantity are fitted independently on two regions. Trials are classed by a no-report marker of perceptual state and matched on inter-regional phase coherence and on each region’s own decoding error, so that agreement cannot follow from single-region fidelity (Eq. (13)). The statistic is agreement in excess of a null that pairs decodes across trials within the same stimulus and error stratum.

756 endpoint is needed to evaluate the interpretation; absence of a report alone
 757 does not define absence of experience. A unified episode with incompati-
 758 ble decoded descriptions would challenge the unity identification. A unified
 759 episode lacking the proposed self-encoding would challenge the minimal-self
 760 claim. Conversely, satisfying the mathematical conditions without such an
 761 episode would challenge their sufficiency as correlates.

762 For the physical route, cortical actuator modes and prices, the geometry–
 763 coupling relation and a dynamical onset model remain to be measured. The
 764 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
 765 diction. Controlled changes in coordination and independently decoded con-

766 tent agreement would address the representational connection. The synthetic
767 decoder exercise calibrates part of these measurements, and the EEG exercise
768 identifies what a pooled bipolar estimand cannot measure; neither identifies
769 a neural content code or tests experience.

770 **10.4 Explanatory scope**

771 Integrated-information, global-workspace and predictive-processing accounts
772 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
773 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
774 relations among coupling, local content and reconstruction. Its formal field
775 condition requires a continuous kernel, a normalized substrate, a specified
776 average coupling and noise, and a site-contribution bound. Those conditions
777 do not identify an electromagnetic substrate; the extracellular interpretation
778 needs the separate calibration of Section 9. The joint witness shows that
779 the formal hypotheses can hold together in a declared model. A cortical
780 realization and an experiential interpretation require additional evidence.

781 Four limitations bound every claim above. Synchrony is not itself an
782 experiential measure, and nothing here establishes a consciousness criterion:
783 the conditions are proposed correlates to be tested against report or another
784 independent endpoint. The present EEG analysis checks a measurement
785 pipeline and does not validate any experiential claim. The formal results
786 are conditional on declared encoders, covers and maps whose cortical coun-
787 terparts are unmeasured. The cover on which compatibility is assessed still

788 needs a selection principle, so the unity correlate awaits an operational defi-
789 nition.

790 The remaining explanatory question is whether these relations occur in
791 neural activity and matter for experience. The framework identifies candidate
792 correlates and tests of their physical and representational premises. It does
793 not explain why satisfying them would be accompanied by experience.

794 11 Conclusions

795 Phase coherence, compatibility and self-reconstruction are separable condi-
796 tions. Coherence constrains shared content only through encoders shared on
797 overlaps and with a bound that grows as the square root of the population
798 read. That bound is vacuous beyond 10 sites at the simulated sheet's patch
799 order, so for populations of cortical size the positive link from coherence
800 to content is the bridge assumption E78 and not a derived result. Com-
801 patible local descriptions, not synchrony, determine one global state; and a
802 contracting self-map fixes that state while reconstructing only a family of
803 bounded diameter. Each result replaces an assumption about synchrony by
804 a measurement: independently decoded agreement on overlaps, a declared
805 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
806 ison of Section 10.2 states how such measurements could count against the
807 proposed link to conscious content.

Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material's convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
U_{inst}, κ	Section 3.2	Installed energy of a declared actuator and the mode-data factor bounding its mean coupling by κU_{inst}
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (9)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (7)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time; in Eq. (4), τ is the duration of a change
J, σ_J, C	Eq. (4)	Probability current of the phase density, its current functional and a current budget
T	Section 5	Number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from

Table 2 (continued)

Symbol	Defined	Meaning
$\gamma, \mathcal{T}, f$	Eq. (10)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (9)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (7)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (10)	Field amplitude, in the cortical identification
s, s_*	§4–5	A global section and the one the reconstruction fixes

809 Data and code availability

810 The Lean development, simulation scripts, tests and saved numerical sum-
811 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
812 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-
813 lib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publication
814 simulation values are generated from saved summaries without rerunning

815 production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
816 ds005620 (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,
817 1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
818 and selection are described in Supplemental Material and implemented in
819 `simulations/empirical_collapse.py`. The repository contains the repro-
820 ducible analysis and does not redistribute the raw recordings.

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824 **Use of AI tools**

825 AI assistance to this work was substantial and is not separable by task.
826 Several coding assistants and chat interfaces were used in an interleaved way
827 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
828 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
829 5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
830 Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
831 v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them
832 they contributed to the Lean development, the simulation and analysis code,
833 and the drafting and revision of both documents. Because the tools were

834 used in combination and in alternation, no section, proof, figure or result is
835 attributable to any one of them, and none is claimed to be. The author is
836 responsible for all content, including everything a tool produced. Machine
837 checking establishes the stated Lean results under their hypotheses; it does
838 not validate biological interpretations or replace scientific review of generated
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